

Is a Farmer Still a Farmer? AI Drones and the Cost of Convenience in Vietnam's Rice Basket

Why Are We Talking About This Topic?

The Arrival of AI Drones in the Delta

The Mekong Delta produces around 90 percent of Vietnam's exported rice (World Bank 8), making it the heart of cultural identity and food security here in Vietnam. Since the late 1980s, farmers have dealt with pests by carrying hand sprayers, walking their plots, and applying chemicals based on personal judgment. However, this process changed in 2021, when Lê Quốc Trung, who grew up in a rice-farming family in the delta's Tràm Chim area, partnered with the Chinese agricultural technology company XAG to bring AI-equipped drones into the region (Negrello). The drones have several functions, such as planting seeds and spraying pesticides while farmers can operate them directly or hire professional pilots for these drones (Negrello). Due to its own technological advancements, XAG now estimates that Vietnam has as many as four thousand agricultural drones in operation (Negrello).

What the Technology Can Do

This new drone technology has also proved its own capabilities, with newer systems pairing drones with multispectral cameras and AI models that detect pests and plant stress before visible symptoms appear. More specifically, the pest detection accuracy of these drones has grown to between 88 and 91 percent ("AI in Multispectral Imaging"). Therefore, a drone team can now spray a field in a fraction of the time manual labor requires, while cutting pesticide use by roughly half and water use by up to 70 percent compared to conventional spraying ("Drones Reshape Agriculture").

The statistics explain why farmers are switching to using these drones ("Use of Agricultural Drones"). However, there are several things that farmers have had to give up for this change. To be more specific, farming in Vietnam has never just been a job, but it is also tied to the memory and identity of the Vietnamese people. Therefore, something is lost when the drone replaces the hand sprayer or plants and fertilizes seeds for a farmer. However, the farmers making that trade are still right to make it, and an account of the delta has to hold both at once.

Questioning How the Land Is Perceived

Heidegger on Enframing and Standing Reserve

German philosopher Martin Heidegger warned us that modern technology changes how we view the world around us (Heidegger). He calls this mindset enframing. Under the concept of enframing, we stop perceiving

nature as having its own presence and instead see it as a "standing reserve," which means a stockpile of raw resources waiting to be optimized (Heidegger). For example, a forest is not seen as a forest, but rather a source of wood collection, and a natural river turns into a source of hydroelectric power, instead of being seen as an ecosystem.

When applying this idea to the Mekong Delta, with the traditional methods of farming such as planting the seeds and hand-spraying, the farmer has a physical presence on the farmland. In contrast, an AI drone system replaces the physical presence of the farmer with multispectral cameras and automated flight paths that were programmed by an algorithm. Therefore, when a farmer wants to manage his or her plot, he or she is observing the farmland through a digital grid rather than walking around the plot physically.

From Farmer to Software Operator

This shift creates a major downside. More precisely, when farmers rely on an app to spot pests and direct spraying, they would unintentionally surrender their own role as a farmer. In particular, with the integration of AI drones, farmers would stop using their personal judgment and knowledge, but would follow the instructions of the drone, essentially turning the farmer into a software operator or a customer buying a technical service. Due to the change of roles after using AI drones, the relationship of the farmer to his or her farmland would become significantly thinner, since farmers are no longer taking care of their land physically.

Concern over Knowledge

Berry on Knowledge That Cannot Be Written Down

Wendell Berry offers a different view, questioning the knowledge of a farmer after incorporating technology and AI into farming. Throughout his career, Berry defended local farming. He argued that good farming depends on a farmer's deep and personal knowledge of his or her piece of land that was built over decades and passed down to family members rather than being written down in a manual (Berry).

How Attention Fades When an App Takes Over

From the point of view of Berry, knowing which corner of a field floods first, or how a certain patch of soil smells before rain, comes only from years of attention and care a farmer has for his or her plot of land (Berry). Therefore, when an app eliminates the need for farmers to pay attention to small details of their land, the bond built through farmland starts to fade as well. The reason for this is that a farmer who cares for their farmland only by relying on drones to do their job will eventually spend less time interacting and taking care of the land themselves, leading to the loss of personal judgment about their own crops.

Is a Farmer Still a Farmer?

Once again, this shift directly attacks the identity of the farmer. To be more specific, traditionally, a Vietnamese farmer had significant knowledge about his or her plot due to the frequent physical work on the plot. However, when farmers use AI drones, not only would they work significantly less on the plot, but a

farmer's knowledge would no longer be required. Consequently, a philosophical question arises : "Is a farmer still a farmer when AI completes their job entirely?"

That is the second half of the loss, and it is significantly harder to recover from. This is mainly because while a drone system can be implemented and replaced instantaneously, a farmer's knowledge takes years of experience to master and pass on to the next generation.

Why Farmers Take the Trade

However, not every scholar views the issue the same way. In *The Moral Economy of the Peasant*, James C. Scott studied village communities across Southeast Asia and found that peasant households ran on a subsistence ethic (Scott). Under this ethic, a family judged a new tool or technique on whether it lowered the risk of falling below the level needed to survive a bad season (Scott). As a result, when a farmer chooses to incorporate drones into his or her plot of land, he or she is testing whether drone systems would protect his or her household during a bad season rather than whether it would ruin older traditions. This is mainly because climate change and unpredictable weather are both contributing to shrinking profits ("In Vietnam"), which places pressure on farming households to pay their bills and support their families. For example, fertilizer alone accounts for about 30 percent of the cost of producing rice in the Mekong Delta ("Fertilize Right"). Moreover, while growers bear 60 to 70 percent of total production costs across the rice supply chain, they receive only about 30 percent of the profit (Minh). Therefore, when a solution arises, such as the new drone system that makes a farmer's job easier and more effective, it is likely that a Vietnamese farmer would incorporate these drones into his or her farmland. What Scott adds overall is that this is a serious situation and not just a lapse in judgment. More precisely, a household choosing to adopt the drone is weighing a thinner bond with their farmland against a harvest that survives a bad year, and it is obvious that farmers would prefer the second alternative.

Holding the Views Together

Where the Three Arguments Meet

While there are disagreements, these differences show that different views can coexist. Heidegger warns that technology reshapes perception (Heidegger). Berry warns about lost knowledge (Berry). Scott shows that farmers adapt on their own terms, and that survival sometimes requires change (Scott). This means that a farmer's relationship with his or her land can lose something at the same time his or her family gains something, such as more stable income or fewer lost harvests while losing attention to and connection with his or her farm plot. This is because ethics usually does not offer an answer but forces a person to trade a thinning bond with the land for a steadier income.

Consequently, if a technology helps a farmer produce more rice and earn more money, but slowly changes the farmer's relationship with his or her own land and identity, has that technology improved the farmer's

life? There is no single correct answer to this question; a person's response depends on what he or she cares about most: financial stability or connection and personal knowledge.

Works Cited

- "AI in Multispectral Imaging for Pest Detection." *Drone Spray Pro*, 10 May 2026, dronespraypro.com/blogs/news/ai-multispectral-imaging-pest-detection.
- Associated Press. "In Vietnam, Farmers Cut Methane Emissions by Changing How They Grow Rice." *Inquirer.net*, 23 Apr. 2024, business.inquirer.net/455945/in-vietnam-farmers-cut-methane-emissions-by-changing-how-they-grow-rice.
- Berry, Wendell. *The Unsettling of America: Culture and Agriculture*. Sierra Club Books, 1977.
- "Drones Reshape Agriculture, but Challenges Remain." *Việt Nam News*, 7 May 2026, vietnamnews.vn/society/1780879/drones-reshape-agriculture-but-challenges-remain.html.
- "Fertilize Right Combines Science and Partnership to Transform Vietnam's Rice Production." *International Rice Research Institute*, 19 Mar. 2025, irri.org/news-and-events/news/fertilize-right-combines-science-and-partnership-transform-vietnams-rice.
- Heidegger, Martin. "The Question Concerning Technology." *The Question Concerning Technology and Other Essays*, translated by William Lovitt, Harper & Row, 1977, pp. 3–35. Cornell University, nissenbaum.tech.cornell.edu/papers/heidegger_concerningtechnology.pdf.
- Minh, Le Nguyet. "Rice Miracle or Mirage: Who Gains from High Prices in Vietnam?" *The Politics of Poverty*, Oxfam America, 23 Oct. 2013, politicsofpoverty.oxfamamerica.org/rice-miracle-or-mirage-who-gains-from-high-prices-in-vietnam/.
- Negrello, Chiara. "Rice Farming Gets an AI Upgrade." *Hakai Magazine*, 17 June 2024, hakaimagazine.com/videos-visuals/rice-farming-gets-an-ai-upgrade/.
- Scott, James C. *The Moral Economy of the Peasant: Rebellion and Subsistence in Southeast Asia*. Yale University Press, 1976.
- "Use of Agricultural Drones on the Rise in the Mekong Delta." *Việt Nam News*, 3 Dec. 2024, vietnamnews.vn/society/1688198/use-of-agricultural-drones-on-the-rise-in-the-mekong-delta.html.
- World Bank. *An Overview of Agricultural Pollution in Vietnam: The Crops Sector*. Written by Tin Hong Nguyen, World Bank Group, 2017, documents1.worldbank.org/curated/en/988621516787454307/pdf/122934-WP-P153343-PUBLIC-Vietnam-crops-ENG.pdf.